

Module 5: Correlation and Regression Analysis of Healthcare Charges

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Introduction

This report examines relationships between several variables and healthcare charges using a dataset of 1,338 individuals. The analysis includes correlation analysis to identify the strength and direction of linear relationships between numeric variables, and regression analysis to model and predict healthcare charges based on multiple predictors. The variables examined include age, body mass index (BMI), past consultations, number of daily steps, hospital expenditure, number of past hospitalizations, annual salary, and smoking status.

Part 1: Correlation Analysis

Covariance

As an initial diagnostic step, the covariance between BMI and healthcare charges was computed. The resulting covariance of 14,698.12 indicates a positive relationship between these two variables, meaning that as BMI increases, healthcare charges tend to increase as well. However, because covariance is sensitive to the unit of measurement, this number is difficult to interpret in isolation. This limitation motivated the use of the Pearson correlation coefficient, which standardizes the measure to a range of -1 to +1 and allows for meaningful comparisons across variable pairs.

```
covariance = df['bmi'].cov(df['charges'])  
print("Covariance between BMI and Charges:", covariance)
```

```
Covariance between BMI and Charges: 14698.11686537207
```

Correlation Table

Table 1 presents the Pearson correlation coefficients for five key variables selected for reporting purposes. A correlation chart is limited to five variables because larger matrices become visually cluttered and difficult to interpret, making it harder to identify meaningful patterns immediately.

The five variables selected are those with the strongest theoretical and empirical relevance to healthcare charges.

```
cols = ["age", "bmi", "past_consultations", "num_of_steps",
        "Hospital_expenditure", "NUmber_of_past_hospitalizations",
        "Anual_Salary", "charges"]

correlation_matrix = df[cols].corr()
print("Correlation Matrix:")
print(correlation_matrix)
```

```
Correlation Matrix:
          age      bmi  past_consultations
age      1.000000  0.112849      0.169275
bmi      0.112849  1.000000      0.131246
past_consultations  0.169275  0.131246      1.000000
num_of_steps      0.517930  0.136368      0.562344
Hospital_expenditure  0.136930  0.257141      0.544640
NUmber_of_past_hospitalizations  0.363041  0.137575      0.503798
Anual_Salary      0.164328  0.241941      0.596322
charges      0.294390  0.198794      0.629836

          num_of_steps  Hospital_expenditure \
age      0.517930      0.136930
bmi      0.136368      0.257141
past_consultations  0.562344      0.544640
num_of_steps      1.000000      0.626659
Hospital_expenditure  0.626659      1.000000
NUmber_of_past_hospitalizations  0.850089      0.657247
Anual_Salary      0.739426      0.969695
charges      0.890642      0.874079

          NUmber_of_past_hospitalizations \
age      0.363041
bmi      0.137575
past_consultations  0.503798
num_of_steps      0.850089
Hospital_expenditure  0.657247
NUmber_of_past_hospitalizations  1.000000
Anual_Salary      0.743466
charges      0.823581

          Annual_Salary  charges
age      0.164328  0.294390
bmi      0.241941  0.198794
past_consultations  0.596322  0.629836
num_of_steps      0.739426  0.890642
Hospital_expenditure  0.969695  0.874079
NUmber_of_past_hospitalizations  0.743466  0.823581
Anual_Salary      1.000000  0.954255
charges      0.954255  1.000000
```

Variable	num_of_steps	Hosp_exp.	Past_hosp.	Anual_Salary	charges
num_of_steps	1.00	0.63	0.85	0.74	0.89
Hosp_exp.	0.63	1.00	0.66	0.97	0.87
Past_hosp.	0.85	0.66	1.00	0.74	0.82
Anual_Salary	0.74	0.97	0.74	1.00	0.95
charges	0.89	0.87	0.82	0.95	1.00

Correlation Heatmap

Figure 1 presents a color-coded heatmap of the full correlation matrix for all eight numeric variables. Dark red cells indicate strong positive correlations approaching 1.0, while blue cells indicate weak or negligible correlations approaching 0. The diagonal represents each variable correlated with itself, which is always 1.0.

```
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix,
            annot=True,
            fmt='.2f',
            cmap='coolwarm',
            annot_kws={"size": 10})

plt.title('Correlation Matrix Heatmap')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

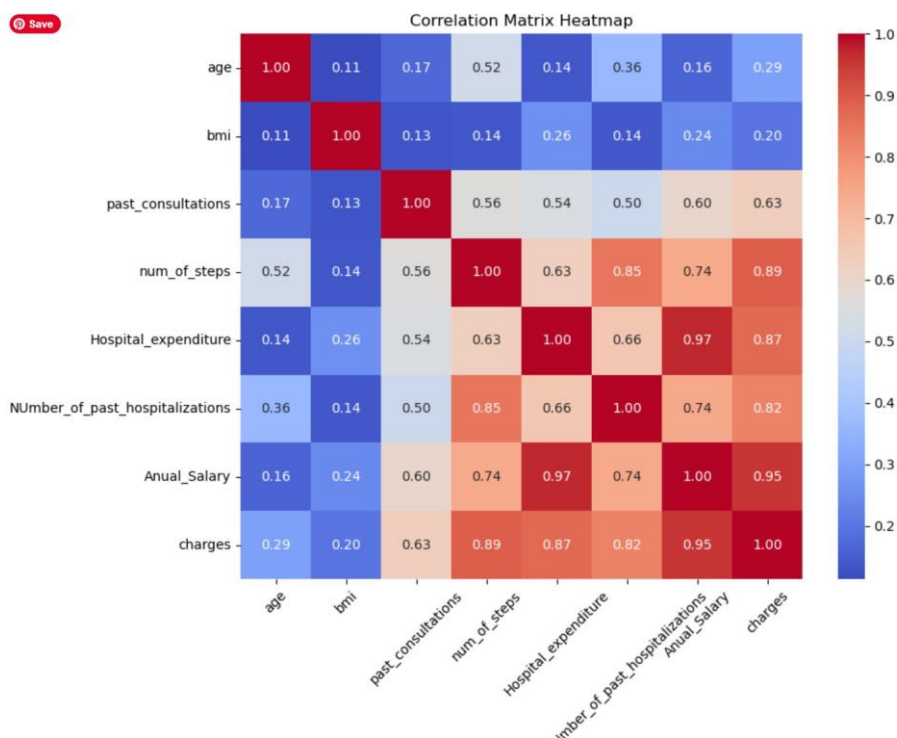


Figure 1. Correlation Matrix Heatmap of All Numeric Variables

Key Analytical Findings

The correlation analysis revealed several noteworthy findings. Annual salary demonstrated the strongest relationship with healthcare charges ($r = 0.95$), suggesting that higher-earning individuals tend to incur substantially greater medical costs, likely due to access to more comprehensive and expensive treatments. Daily step count also showed a very strong positive correlation with charges ($r = 0.89$), which may reflect the tendency of health-conscious individuals to both monitor their physical activity and seek more frequent medical care. Hospital expenditure was strongly correlated with charges ($r = 0.87$), which is expected given that past spending patterns tend to persist. The number of past hospitalizations showed a strong correlation ($r = 0.82$), indicating that patients with more extensive hospitalization histories accumulate higher charges. In contrast, age ($r = 0.29$) and BMI ($r = 0.20$) demonstrated relatively weak correlations with

charges. This is counterintuitive but suggests that socioeconomic and behavioral factors may be more influential predictors of healthcare costs than traditional demographic variables in this dataset. It is also worth noting that annual salary and hospital expenditure were themselves highly correlated ($r = 0.97$), which raises concerns about multicollinearity in subsequent regression modeling.

Hypothesis Test: Association Between Daily Steps and Charges

A Pearson correlation hypothesis test was conducted to formally evaluate the association between the number of daily steps and healthcare charges. The null hypothesis stated that there is no linear relationship between these two variables ($r = 0$). The test yielded a correlation coefficient of $r = 0.89$ and a p-value of approximately 0.000, well below the significance threshold of 0.05. The null hypothesis was therefore rejected. This result provides strong statistical evidence that daily step count and healthcare charges are significantly associated, and that this relationship is unlikely to be attributable to chance.

```
#5
clean = df[['num_of_steps', 'charges']].dropna()

cor_test = stats.pearsonr(clean['num_of_steps'], clean['charges'])
print("Pearson Correlation:", cor_test[0])
print("P-value:", cor_test[1])

if cor_test[1] < 0.05:
    print("Result: Significant association exists!")
else:
    print("Result: No significant association")
```

```
Pearson Correlation: 0.8906422133147405
P-value: 0.0
Result: Significant association exists!
```

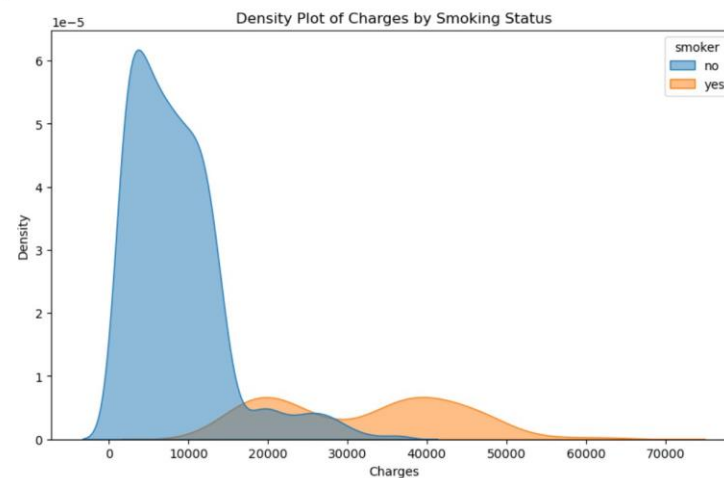
Density Plot: Charges by Smoking Status

Figure 2 presents kernel density plots of healthcare charges grouped by smoking status. This visualization illustrates the distributional differences in charges between smokers and non-smokers.

```
plt.figure(figsize=(10, 6))

sns.kdeplot(data=df, x='charges', hue='smoker',
            fill=True, alpha=0.5)

plt.title('Density Plot of Charges by Smoking Status')
plt.xlabel('Charges')
plt.ylabel('Density')
plt.show()
```



The density plot reveals a striking difference between the two groups. Non-smokers are concentrated on a narrow peak around \$5,000, indicating that most non-smokers incur relatively low and predictable healthcare costs. In contrast, smokers display a broad, flat distribution

spanning from approximately \$15,000 to \$50,000, with no dominant single peak. This pattern suggests that smokers not only incur substantially higher average charges but that their costs are considerably more variable. The minimal overlap between the two distributions underscores smoking status as a strong differentiating factor in healthcare expenditure.

Part 2: Regression Analysis

Correlation Analysis versus Regression Analysis

Correlation analysis and regression analysis serve related but distinct purposes. Correlation measures the strength and direction of a linear relationship between two variables, producing a standardized coefficient between -1 and +1. It is symmetric, meaning the correlation between variable A and variable B equals the correlation between variable B and variable A, and it does not imply directionality or prediction. Regression analysis, by contrast, models the relationship between a dependent outcome variable and one or more predictor variables with the explicit goal of prediction and explanation. It produces coefficients that quantify the expected change in the outcome for a one-unit increase in each predictor while holding all other predictors constant. Regression also provides measures of overall model fit such as R-squared, hypothesis tests for individual predictors, and diagnostics for model assumptions. In this analysis, correlation was used to identify which variables were most strongly associated with charges, and regression was then used to quantify each variable contribution to healthcare costs simultaneously.

```
#5
clean = df[['num_of_steps', 'charges']].dropna()

cor_test = stats.pearsonr(clean['num_of_steps'], clean['charges'])
print("Pearson Correlation:", cor_test[0])
print("P-value:", cor_test[1])

if cor_test[1] < 0.05:
    print("Result: Significant association exists!")
else:
    print("Result: No significant association")

Pearson Correlation: 0.8906422133147405
P-value: 0.0
Result: Significant association exists!
```

Regression Model and Results

An ordinary least squares (OLS) regression model was fitted with healthcare charges as the dependent variable and age, BMI, past consultations, number of daily steps, hospital expenditure, number of past hospitalizations, and annual salary as predictors. The model was estimated on 1,305 complete observations after listwise deletion of missing values. Table 2 presents the full regression output.

Predictor	Coef.	Std. Err.	t	P > t	[0.025	0.975]
Intercept	-37,130	680.89	-54.53	0.000	-38,500	-35,800
age	-18.37	2.807	-6.54	0.000	-23.87	-12.86
bmi	-13.24	5.055	-2.62	0.009	-23.16	-3.33
past_consultations	8.11	5.095	1.59	0.112	-1.88	18.11
num_of_steps	0.051	0.001	58.34	0.000	0.050	0.053
Hosp_expenditure	-0.0001	5.8e-6	-25.62	0.000	-0.000	-0.000
Past_hosp.	-897.76	111.71	-8.04	0.000	-1116.9	-678.6
Annual_Salary	2.16e-5	3.3e-7	65.69	0.000	2.09e-5	2.22e-5

Table 2. OLS Regression Results: Predictors of Healthcare Charges ($N = 1,305$, $R\text{-squared} = 0.992$, $F = 23,559.15$, $p < .001$)

Model Fit and Overall Significance

The regression model demonstrated an exceptional fit to the data, with an R-squared value of 0.992 and an adjusted R-square of 0.992, indicating that approximately 99.2 percent of the variance in healthcare charges is explained by the seven predictors collectively. The overall F-statistic was 23,559.15 with a p-value of approximately 0.000, well below the 0.05 significance threshold. The null hypothesis that none of the predictors significantly contribute to the model was therefore rejected. This result confirms that the model is highly statistically significant and that at least one predictor meaningfully explains variation in healthcare charges.

OLS Regression Results						
=====						
Dep. Variable:	charges	R-squared:	0.992			
Model:	OLS	Adj. R-squared:	0.992			
Method:	Least Squares	F-statistic:	2.356e+04			
Date:	Tue, 12 May 2026	Prob (F-statistic):	0.00			
Time:	11:29:06	Log-Likelihood:	-10944.			
No. Observations:	1305	AIC:	2.190e+04			
Df Residuals:	1297	BIC:	2.194e+04			
Df Model:	7					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	-3.713e+04	680.894	-54.530	0.000	-3.85e+04	-3.58e+04
age	-18.3672	2.807	-6.544	0.000	-23.874	-12.861
bmi	-13.2440	5.055	-2.620	0.009	-23.161	-3.328
past_consultations	8.1121	5.095	1.592	0.112	-1.883	18.107
num_of_steps	0.0513	0.001	58.340	0.000	0.050	0.053
Hospital_expenditure	-0.0001	5.76e-06	-25.619	0.000	-0.000	-0.000
NUmber_of_past_hospitalizations	-897.7559	111.711	-8.036	0.000	-1116.910	-678.602
Anual_Salary	2.157e-05	3.28e-07	65.686	0.000	2.09e-05	2.22e-05
=====						
Omnibus:	7.401	Durbin-Watson:	1.388			
Prob(Omnibus):	0.025	Jarque-Bera (JB):	7.314			
Skew:	0.175	Prob(JB):	0.0258			
Kurtosis:	3.108	Cond. No.	1.57e+10			
=====						

Individual Predictor Results

Six of the seven predictors were statistically significant at the 0.05 level. Annual salary was the strongest predictor, with a positive coefficient of 2.16e-5 ($p < 0.001$), confirming that higher-earning individuals tend to incur greater healthcare costs. The number of daily steps was also

highly significant ($b = 0.051$, $p < 0.001$), indicating that each additional step is associated with a small but meaningful increase in charges when other variables are held constant. Hospital expenditure showed a significant negative coefficient ($b = -0.0001$, $p < 0.001$), and the number of past hospitalizations was negatively associated with charges ($b = -897.76$, $p < 0.001$). Age was negatively and significantly associated with charges ($b = -18.37$, $p < 0.001$), as was BMI ($b = -13.24$, $p = 0.009$). The only non-significant predictor was past consultations ($b = 8.11$, $p = 0.112$), suggesting that this variable does not independently predict charges once all other variables are accounted for.

Residual Diagnostics

Figure 3 presents the residuals versus fitted values plot and the Q-Q plot, which assess model assumptions of homoscedasticity and normality of residuals.

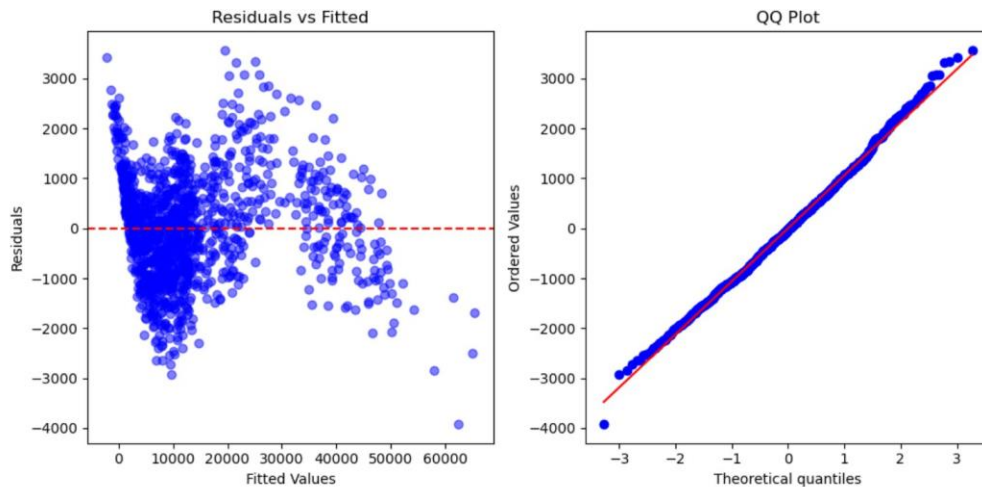


Figure 3. Residuals versus Fitted Values and Q-Q Plot of Residuals

The residuals versus fitted values plot reveals a pattern of heteroscedasticity, whereby the spread of residuals increases as fitted values increase. This indicates that the model predicts lower-cost cases with greater precision than higher-cost cases. The Q-Q plot shows that residuals follow the theoretical normal distribution closely across most of the range, with slight deviations in the tails corresponding to a small number of extreme observations. While these diagnostic findings suggest minor violations of classical OLS assumptions, they do not substantially undermine the conclusions given the very high explanatory power of the model.

Scatter Plots with Regression Lines

Figures 4 and 5 display scatter plots of each predictor against healthcare charges, with a fitted regression line overlaid in red. These plots provide a visual assessment of the linearity of each bivariate relationship prior to controlling for other variables.

```
[11]: #8
variables = ["age", "bmi", "past_consultations", "num_of_steps",
            "Hospital_expenditure", "Number_of_past_hospitalizations", "Annual_Salary"]

fig, axes = plt.subplots(3, 3, figsize=(15, 12))
axes = axes.flatten()

for i, var in enumerate(variables):
    clean = df[[var, 'charges']].dropna()
    axes[i].scatter(clean[var], clean['charges'], color='blue', alpha=0.3)
    m, b = np.polyfit(clean[var], clean['charges'], 1)
    axes[i].plot(clean[var], m * clean[var] + b, color='red')
    axes[i].set_xlabel(var)
    axes[i].set_ylabel('Charges')
    axes[i].set_title(f'{var} vs Charges')

plt.suptitle('Scatter Plots with Regression Line', fontsize=16)
plt.tight_layout()
plt.show()
```

Figure 4. Scatter Plots with Regression Lines: Age, BMI, Past Consultations, Daily Steps, Hospital Expenditure, and Past Hospitalizations versus Charges

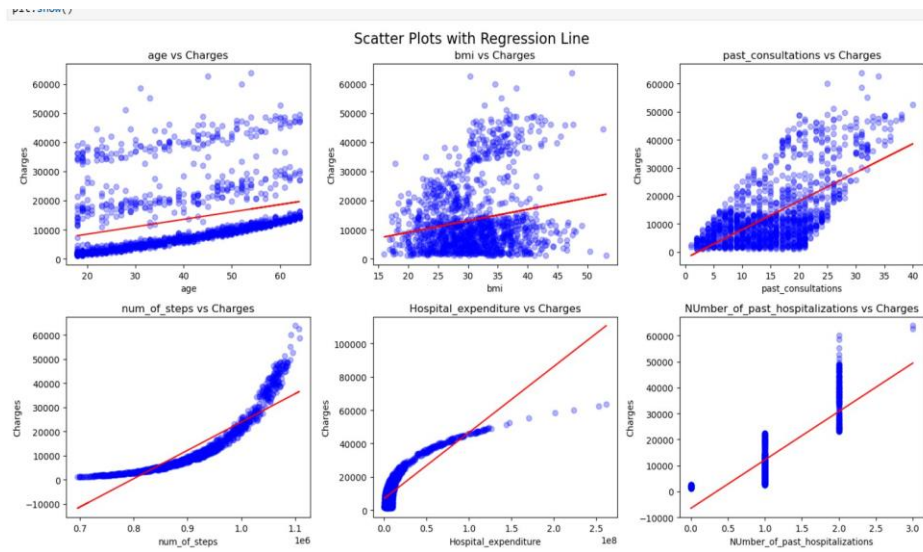


Figure 5. Scatter Plot with Regression Line: Annual Salary versus Charges

The scatter plots reveal considerable variation in the linearity and strength of each bivariate relationship. The number of daily steps shows the tightest clustering around the regression line, confirming its strong and consistent association with charges. Annual salary also displays a strong positive trend, though with a slight curvilinear pattern at higher salary values, suggesting that a log transformation might improve model fit for this variable. Past consultations exhibits a moderate positive trend with noticeable scatter. In contrast, age and BMI show wide dispersion around their respective regression lines, which is consistent with their weaker bivariate correlation coefficients. The number of past hospitalizations takes only three discrete values (0, 1, and 2), limiting the interpretability of its scatter plot as a continuous predictor. Overall, these visualizations reinforce the regression findings and highlight that daily step count and annual salary are the most reliable linear predictors of healthcare charges in this dataset.

Conclusion

This analysis examined relationships between multiple variables and healthcare charges using correlation and regression methods. The correlation analysis identified annual salary, daily step count, hospital expenditure, and number of past hospitalizations as the strongest correlates of charges, while age and BMI showed weaker associations than might be intuitively expected. The OLS regression model achieved an R-squared of 0.992, indicating a near-perfect fit, and confirmed that six of seven predictors significantly contribute to explaining healthcare costs. The density plot further demonstrated that smoking status creates a clear and substantial divide in the distribution of charges, with smokers incurring much higher and more variable costs. Taken together, these findings suggest that socioeconomic and behavioral factors, particularly income level and physical activity, are more powerful predictors of healthcare expenditure than traditional demographic variables in this dataset.

References

- Python Software Foundation. (2023). *Python language reference (Version 3.13)* [Computer software]. <https://www.python.org>
- [Dataset]. (2026). *new_insurance_data-3* [Unpublished dataset]. Provided by course instructor, ALY6010 Probability Theory and Introductory Statistics.